

16.3.1 Sufficient Condition for Extreme Values

A function $y = f(x_1, x_2)$ is said to be at maximum at a point (a, b) if $f(a, b) > f(x_1, x_2)$ for all values of (x_1, x_2) in the domain $a - \delta < x_1 < a + \delta$ and $b - \delta < x_2 < b + \delta$. Here, $x_1 \neq a$, $x_2 \neq b$ and δ is a very small quantity, i.e.,

$$f(a + h, b + k) - f(a, b) < 0$$

where, $|h| < \delta$ and $|k| < \delta$.

This implies, the function $y = f(x_1, x_2)$ is said to have a maximum at the point (a, b) if any movement from the point in any direction gives us a lower value of y .

If $dy = 0$ at the point of maximum and $dy < 0$ elsewhere, we can conclude that the sufficient condition to have a maximum is that the change in dy should be diminishing i.e.,

$$d(dy) < 0 \Rightarrow d^2y < 0.$$

From $y = f(x_1, x_2)$ we have

$$dy = f_1 dx_1 + f_2 dx_2 = 0$$

$$\text{or, } d^2y = d(f_1 dx_1 + f_2 dx_2)$$

$$\text{i.e., } d^2y = d(f_1 dx_1) + d(f_2 dx_2)$$

$$= df_1(dx_1) + df_2(dx_2)$$

$$= (f_{11} dx_1 + f_{12} dx_2) dx_1 + (f_{21} dx_1 + f_{22} dx_2) dx_2$$

By Young's theorem, $f_{12} = f_{21}$.

Thus, $d^2y < 0$ implies

$$f_{11} (dx_1)^2 + 2f_{12} dx_1 dx_2 + f_{22} (dx_2)^2 < 0 \quad \text{----- (i)}$$

Note that f_{11} , f_{12} and f_{22} are derived on any particular point, say (a, b) . Thus, their values can be assumed to be constant. However, dx_1 and dx_2 are variables.

Let, $u = dx_1$, $v = dx_2$, $a = f_{11}$, $b = f_{22}$, $h = f_{12}$.

The left-hand expression of (i) can be written in a quadratic form as

$$q = au^2 + 2huv + bv^2$$

$$= a \left(u^2 + \frac{2h}{a} uv + \frac{h^2}{a^2} v^2 \right) + \left(b - \frac{h^2}{a} \right) v^2$$

$$= a \left(u + \frac{h}{a} v \right)^2 + \frac{ab - h^2}{a} (v^2).$$

Since both $\left(u + \frac{h}{a} v \right)^2$ and v^2 are positive, sign of q depends entirely on the

values of the coefficients a , b and h . The inequality $q < 0$ holds if $a < 0$ and

$$ab - h^2 > 0.$$

Again, since the product a b must be positive in order to get positive $(ab-h^2)$, both a and b must bear the same sign. Thus, if $a < 0$, $b < 0$ and $ab - h^2 > 0$, then $q < 0$.

Alternatively, $d^2y < 0$, if $f_{11} < 0$, $f_{22} < 0$ and $f_{11} f_{22} - f_{12}^2 > 0$. This is the sufficient condition for a maximum.

The sufficient condition for a minimum of a bivariate function is

$f_{11} > 0$, $f_{22} > 0$, $f_{11} f_{22} - f_{12}^2 > 0$ which can be derived in a similar way.

Let us now consider the case when $f_{11} f_{22} < (f_{12})^2$. A stationary point of a bivariate function, where $f_1 = 0 = f_2$ is called

i) A saddle point if

$f_{11} f_{22} < f_{12}^2$ and f_{11} and f_{22} have different signs.

ii) An inflexion point if $f_{11} f_{22} < f_{12}^2$ and f_{11} , f_{22} have the same sign.

If $f_{11} f_{22} = f_{12}^2$, the change becomes inconclusive.

Now, let us consider the case of two variables and take the following expression:

$$q = au^2 + 2huv + bv^2.$$

$$= au^2 + huv + huv + bv^2.$$

This may be put in the following matrix form:

$$q = \begin{bmatrix} u & v \end{bmatrix} \begin{bmatrix} a & h \\ h & b \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix}$$

Since we are not concerned with the value of u and v , matrix of order 2×2 i.e.,

$\begin{bmatrix} a & h \\ h & b \end{bmatrix}$ supplies us the required conditions for q to be positive or negative.

i) q will have positive sign

$$\text{if } |a| > 0 \text{ and } \begin{vmatrix} a & h \\ h & b \end{vmatrix} > 0$$

ii) q will have negative sign

$$\text{if } |a| < 0 \text{ and } \begin{vmatrix} a & h \\ h & b \end{vmatrix} > 0$$

Next, consider the position of $|a|$ and $\begin{vmatrix} a & h \\ h & b \end{vmatrix}$.

The determinant $|a|$ is the sub-determinant of H that consists of the first element of the principal diagonal and is called the first principal minor of H .

The determinant $\begin{vmatrix} a & h \\ h & b \end{vmatrix}$ can also be considered as sub-determinant of H . Since it

involves the first and second elements on the principal diagonal of H , it is called

the second principal minor of H . In the case of a function of two variables, only two principal minors are available but there will be three if the function involves three variables.

Check Your Progress 3

- 1) Find the stationary values and test whether they are maximum or minimum for $z = 3x^2 + 6xy + 7y^2$

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- 2) Show that the only stationary point of the function $z = 4x^2 - 2y^2 + 7xy$ is a saddle point.

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16.3.2 Function of Three Variables and Extrema

Assume, $z = f(x, y, w)$.

Then $dz = f_x dx + f_y dy + f_w dw$.

$$d^2z = f_{xx}dx^2 + f_{xy}dxdy + f_{xw}dxdw + f_{yy}(dy)^2 + f_{yw}dydw + f_{wx}dwdx + f_{wy}dw dy + f_{yx}dydx + f_{ww}(dw)^2.$$

$$= \begin{bmatrix} dx & dy & dw \end{bmatrix} \begin{bmatrix} f_{xx} & f_{xy} & f_{xw} \\ f_{yx} & f_{yy} & f_{yw} \\ f_{wx} & f_{wy} & f_{ww} \end{bmatrix} \begin{bmatrix} dx \\ dy \\ dw \end{bmatrix}$$

The determinant of the matrix $\begin{bmatrix} f_{xx} & f_{xy} & f_{xw} \\ f_{yx} & f_{yy} & f_{yw} \\ f_{wx} & f_{wy} & f_{ww} \end{bmatrix}$

shall have three principal minors namely,

$$D_1 = |f_{xx}|, |D_2| = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix},$$

$$\text{and } |D_3| = \begin{vmatrix} f_{xx} & f_{xy} & f_{xw} \\ f_{yx} & f_{yy} & f_{yw} \\ f_{wx} & f_{wy} & f_{ww} \end{vmatrix}$$

Our problem is to determine the restrictions that should be placed upon D_1 , D_2 and D_3 , so that, for any value of dx and dy , the value of d^2z will turn out to be positive or negative.

For this, we once again put the quadratic form of three variables in a way that the three variables always appear as components of some square.

$$\text{Let } q = ax^2 + by^2 + cw^2 + 2hxy + 2fyx + 2gwx$$

$$d^2z = f_{xx}(dx^2) + f_{yy}(dy^2) + f_{ww}(dw)^2 + 2f_{xy} dx dy + 2f_{yw} dy dw + 2f_{xw} dx dw.$$

We can rewrite q in the following form:

$$q = a \left(x + \frac{h}{a}y + \frac{g}{a}w \right)^2 + \frac{ab-h^2}{a} \left(y + \frac{af-gh}{ab-h^2} \right)^2 + \left(\frac{abc-af^2-bg^2-ch^2+2fgh}{ab-h^2} \right) w^2$$

So that

i) for q to be positive, the following must hold:

$$a > 0, ab - h^2 > 0, \text{ and}$$

$$abc - af^2 - bg^2 - ch^2 + 2fgh > 0.$$

The above three conditions can be put as the three principal minors of the matrix

$$\begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}$$

$$D_1 > 0, D_2 > 0 \text{ and } D_3 > 0.$$

Translating these elements in terms of partial derivatives, we have the following three conditions for d^2z to be positive for the given function to have minimum value.

$$D_1 = |f_{xx}| > 0, D_2 = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} > 0,$$

$$D_3 = \begin{vmatrix} f_{xx} & f_{xy} & f_{xw} \\ f_{yx} & f_{yy} & f_{yw} \\ f_{wx} & f_{wy} & f_{ww} \end{vmatrix} > 0.$$

ii) for q to be negative, following conditions must hold:

$$D_1 = a < 0, D_2 = \begin{vmatrix} a & h \\ h & b \end{vmatrix} > 0,$$

$$D_3 = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} < 0.$$

or, in terms of partial derivatives for d^2z to be negative (or the given function to have maximum value), following conditions must hold good:

$$D_1 = |f_{xx}| < 0, D_2 = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} > 0,$$

$$D_3 = \begin{vmatrix} f_{xx} & f_{xy} & f_{xw} \\ f_{yx} & f_{yy} & f_{yw} \\ f_{wx} & f_{wy} & f_{ww} \end{vmatrix} < 0.$$

i.e., the principal minors of the determinant of d^2z should be successively negative, positive and negative.

16.3.3 Function of n Variables and Extrema

When there are n independent variables, the objective function may be expressed as

$$z = f(x_1, x_2, \dots, x_n).$$

The total differential will then be.

$$dz = f_1 dx_1 + \dots + f_n dx_n$$

so that F.O.C. for minimum or maximum will be that all the n first order partial derivatives must vanish (since $dz = 0$) and therefore,

$$f_1 = f_2 = \dots = f_n = 0.$$

For the second order condition

- i) d^2z will have positive sign (i.e., function will have the minimum value) if all the n principal minors of the Hessian determinant:

$$|H| = \begin{vmatrix} f_{11} & \dots & f_{1n} \\ f_{21} & f_{22} & f_{2n} \\ \dots & \dots & \dots \\ f_{n1} & f_{n2} & f_{nn} \end{vmatrix} \text{ are positive.}$$

- ii) d^2z will have negative sign (i.e., function will have maximum value, if n principal minors of H possess alternate signs - the first being negative.

Let us now summarize the extremum conditions in the tabulated form:

For $z = f(x_1, x_2, \dots, x_n)$ to be

Condition	Maximum	Minimum
F.O.C.	$dz = 0$ i.e., $f_1 = f_2 = \dots = f_n = 0$	$dz = 0$ i.e., $f_1 = f_2 = \dots = f_n = 0$
S.O.C	$d^2z < 0$, i.e., $ H_1 < 0$, $ H_2 > 0$, $ H_3 < 0$, --	$d^2z > 0$ i.e., $ H_1 > 0$, $ H_2 > 0$, $ H_3 > 0$, ---

Example 16.4:

- i) Indicate whether the function

$Z = f(x_1, x_2, x_3) = x_1^2 - 3x_1 x_2 + 3x_2^2 + 4x_2 x_3 + 6x_3^2$ has maximum or minimum value.

F.O.C.

$$f_1 = 2x_1 - 3x_2 = 0$$

$$f_2 = -3x_1 + 6x_2 + 4x_3 = 0.$$

$$f_3 = 4x_2 + 12x_3 = 0.$$

Solving these equations, we get $x_1=0$, $x_2=0$ and $x_3=0$.

$$|H| = \begin{vmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{vmatrix} = \begin{vmatrix} 2 & -3 & 0 \\ -3 & 6 & 4 \\ 0 & 4 & 12 \end{vmatrix}$$

for $(x_1, x_2, x_3) = (0, 0, 0)$.

where $f_{ij} = f_{x_i x_j} \forall i, j = 1, 2, 4$.

$$|H_1| = 2 > 0, |H_2| = \begin{vmatrix} 2 & -3 \\ -3 & 6 \end{vmatrix} > 0$$

$$\text{and } |H_3| = \begin{vmatrix} 2 & -3 & 0 \\ -3 & 6 & 4 \\ 0 & 4 & 12 \end{vmatrix} > 0.$$

i.e., all principal minors are positive. Hence, the given function has minimum value equal to zero at point $(0, 0, 0)$.

ii) Determine whether the function $v = f(x, y, z)$

$= -x^3 + 3xz + 2y - y^2 - 3z^2$ has maximum or minimum value.

$$\text{F.O.C: } f_1 = -3x^2 + 3z = 0$$

$$f_2 = 2 - 2y = 0$$

$$f_3 = 3x - 6z = 0$$

Solving these equations, we get $x = 0$, or $1/2$; $y = 1$; $z = 0$ or $1/4$.

Thus, there are two stationary points $(0, 1, 0)$ and $(1/2, 1, 1/4)$ and we have to examine whether the given function has maximum or minimum value at those points.

$$\text{S.O.C: } |H| = \begin{vmatrix} f_{xx} & f_{xy} & f_{xz} \\ f_{yx} & f_{yy} & f_{yz} \\ f_{zx} & f_{zy} & f_{zz} \end{vmatrix} = \begin{vmatrix} -6x & 0 & 3 \\ 0 & -2 & 0 \\ 3 & 0 & -6 \end{vmatrix}$$

At $(0, 1, 0)$

$$|H| = \begin{vmatrix} 0 & 0 & 3 \\ 0 & -2 & 0 \\ 3 & 0 & -6 \end{vmatrix}$$

$$f_{xx} = 0.$$

Thus, S.O.C. cannot be fulfilled.

At $(1/2, 1, 1/4)$,

$$|H| = \begin{vmatrix} -3 & 0 & 3 \\ 0 & -2 & 0 \\ 3 & 0 & -6 \end{vmatrix}$$

$$|H_1| = -3 < 0, |H_2| = 6 > 0, |H_3| = -18 < 0$$

which alternate in sign.

Hence at $(1/2, 1, 1/4)$, the given function possesses maximum value which will be $\frac{17}{16}$.

Check Your Progress 4

1) Examine the following functions for maxima or minima.

a) $z = 2x^2 + xy + 4y^2 + xz + z^2 + 2$

b) $z = e^{2x} - e^y + e^{z^2} - 2(x + e) + y$.

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16.4 LET US SUM UP

In this unit, we have discussed the extrema of a function and the condition under which it attains (relative/ absolute) extrema. We started with single variable case for simplicity extended the argument for multivariable cases. The emphasis was on the technique of how to obtain extrema of a function, which is differentiable.

16.5 KEY WORDS

Local (relative) Maxima and Global (absolute) Maxima : A function attains local maximum at any particular point in the domain of definition implies that the value of the function is maximum in the neighborhood of that point only, and the function can assume higher values elsewhere. This can be non-unique. The global maximum is the highest value of the function with reference to its entire domain and hence unique

Local (relative) Minima and Global (absolute) Minima : A function attains local minimum at any particular point in the domain of definition implies that the value of the function is minimum in the neighborhood of that point only, and the function can assume lower values elsewhere. This can be non-unique. The global minimum is the lowest value of the function with reference to its entire domain and hence unique.

- Points of Inflexion** : The point of inflexion is defined as a point at which a curve changes its curvature. The sufficient condition for a point of inflexion is $f''(x) = 0$ and $f'''(x) \neq 0$. Thus, if a function has $f'(x) = 0$, $f''(x) = 0$ and $f'''(x) \neq 0$ at the point x , the point is said to be stationary and inflexional.
- Stationary Points** : The points, at which first order derivatives are zero, are called stationary points. The values of the function at those points are called stationary values.

16.6 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) $f'(x) = 3(x^2 - 4x + 8) = 3\{(x-2)^2 + 4\} > 0 \quad \forall x$. Thus, the function does not have a maximum or a minimum.
- 2) $x = -1$ and $x = 1$ are minima. $x = 0$ is the maximum

Check Your Progress 2

- 1) $x = -1/3$ is the point of inflexion.

Check Your Progress 3

- 1) Here, $f(x) = 6x + 6y$, $f(y) = 6x + 14y$.
 $f_{xx} = 6$, $f_{xy} = 6$, $f_{yy} = 14$.

F.O.C. requires $f_x = f_y = 0$ i.e.,

$$6x + 6y = 0 \dots\dots\dots(i)$$

$$6x + 14y = 0 \dots\dots\dots(ii)$$

Solving (i) and (ii) for x and y we get, $x = 0$, $y = 0$.

The given function reaches its minimum value at the stationary point and its minimum value is zero. This is because

$$f_{xx}\bigg|_{\substack{x=0 \\ y=0}} = 6 > 0, f_{yy}\bigg|_{\substack{x=0 \\ y=0}} = 14 > 0.$$

$$\text{Also, } f_{xx}.f_{yy} - (f_{xy})^2 \bigg|_{\substack{x=0 \\ y=0}}$$

$$= 84 - 36 = 48 > 0.$$

- 2) Here $f_x = 8x + 7y$, $f_y = -4y + 7x$, $f_{xx} = 8$, $f_{xy} = 7$, $f_{yy} = -4$

F.O.C. for having stationary values is $f_x = f_y = 0$

$$\text{i.e., } 8x + 7y = 0$$

$$-4y + 7x = 0$$

Solution of these equations gives us $x=0$, $y = 0$. Thus, the given function has only one stationary point at $(0, 0)$. Now f_{xx} and f_{yy} have opposite signs and $f_{xx} \cdot f_{yy} - (f_{xy})^2 < 0$. Thus, the stationary point for the given function is a saddle point.

Check Your Progress 4

- 1) a) No maxima or minima
b) Maximum at $(0,0,0)$

16.7 EXERCISES

Q1. Find the maxima and minima of $1 + 2 \sin x + 3 \cos^2 x$ where $\left(0 \leq x \leq \frac{\pi}{2}\right)$.

Ans. To find the maxima and minima of $1 + 2 \sin x + 3 \cos^2 x$.

$\left(0 \leq x \leq \frac{\pi}{2}\right)$ get

$$f(x) = 1 + 2 \sin x + 3 \cos^2 x$$

$$f'(x) = 2 \cos x - 6 \cos x \sin x$$

$$f'(x) = 0 \text{ when } 2 \cos x(1 - 3 \sin x) = 0.$$

i.e., when $\cos x = 0$ and when $\sin x = \frac{1}{3}$.

$$f''(x) = -2 \sin x - 6(\cos^2 x - \sin^2 x)$$

$$\text{when } \cos x = 0, x = \frac{\pi}{2} \therefore \sin x = 1$$

$$\therefore f''(x) = -2 + 6 = 4 > 0.$$

For $\cos x = 0$, $f(x)$ is a minimum and the minimum value is 3.

$$\text{When } \sin x = \frac{1}{3}$$

$$f''(x) = -2 \sin x - 6(1 - 2 \sin^2 x)$$

$$f''(x) \Big|_{\sin x = \frac{1}{3}} = -\frac{2}{3} - 6\left(1 - \frac{2}{9}\right) < 0$$

Therefore, for $\sin x = \frac{1}{3}$, $f(x)$ is a maximum and the maximum value is $4\frac{1}{3}$.

Q2. If $y = x_1^3 + 2x_2^3 - 6x_1^2 + 9x_2^2 - 63x_1 - 60x_2 - 24$, find the optimum points and verify the nature for the function.

Ans. $x_1 = -3$, $x_2 = 2$ saddle point

$x_1 = 7$, $x_2 = -5$ saddle point

$x_1 = -3$, $x_2 = -5$ relative maximum

$x_1 = 7$, $x_2 = 2$ relative minimum

Q3. Examine $y = -x_1^3 + 9x_1 - 4x_2^2$ for relative extrema.

Ans. The necessary conditions for maximum or minimum are

$$f_1 = -3x_1^2 + 9 = 0 \dots(i)$$

$$f_2 = -8x_2 = 0 \dots(ii)$$

from (i) and (ii) we get $x_1 = \pm \sqrt{3}$, $x_2 = 0$.

Thus, we get two stationary points for this function, namely, $(\sqrt{3}, 0)$ and $(-\sqrt{3}, 0)$. At stationary point, $(\sqrt{3}, 0)$ the function has a maximum. At $(-\sqrt{3}, 0)$ the function has a saddle point.



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